

02LQD Specification and Simulation of Digital Systems

2020/06/15

1. Combinational circuit design (3 points)

A car has a low-tire-pressure sensor that outputs the current tire pressure as a 5-bit binary number P . Create a circuit that illuminates a “low tire pressure” indicator light by setting the T output to 1 when the tire pressure P drops below 16.

Design in VHDL a combinational circuit implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (12 points)

Build an electronic combination lock with a reset line rst , a serial input line x and an unlock output u . The unlock output u is at 1 if and only if the sequence 01011 has appeared on x in the last 5 clock ticks. The combination lock is a Moore machine with asynchronous reset.

- Draw the state diagram (4 points)
- Design the FSM resorting to the VHDL behavioural description style with 2 processes (4 points).
- Design the FSM resorting in Verilog with 2 always blocks (4 points).

3. FSM-D design (15 points)

Design a FSM-D to count the number of events on a 1-bit input line B . An event is any change of B from 0 to 1 or from 1 to 0. The number of events is output on a 16-bit output C . C rolls over (to 0) whenever its maximum value is reached. Reset is asynchronous.

Steps:

1. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (7 points)
2. Design a FSM-D (8 points):
 - create 2 processes for behavioural datapath description from the HLSM state diagram
 - create 2 processes for behavioural controller description from the HLSM state diagram
 - connect datapath processes to controller processes to create a single architecture.

Exam rules:

- upload on Portale della didattica in section Elaborati a .zip archive including:
 - running VHDL code and testbench
 - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam

no later than Thursday June 18 11.59 pm

- format of .zip archive name surname_ID (example Rossi_244123)
- remember: no upload $\Rightarrow$ no correction $\Rightarrow$ no oral exam
- once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.